
MATH 525

Probability

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0.1 Intro

Recommended Reading: Probability and Random Processes (Grimmett/Stirzker)

Mathematical study of randomness (random $\leftrightarrow$ deterministic)

How do we model Randomness?

Coin: $P(H) = P(T) = 0.5$

Biased Coin: $P(H) = 0.7, P(T) = 0.3$

Dice: $P(1) = P(2) = \dots = P(6) = \frac{1}{6}$

$P(1 \text{ or } 3) = P(\{1, 3\}) = P(1) + P(3) = \frac{1}{3}$

Power Law: $P(n) = \frac{6}{\pi^2} \frac{1}{n^2}$

$P(n \text{ is even}) = P(\{2, 4, 6, \dots\}) = \frac{1}{2}$

$P(4 \text{ divides } n) = P(\{4, 8, \dots\})$

Sample space and probability measure

Definition 0.1.1 (Sample Space). Let Ω be a set. Let F be a collection of subsets of Ω , which is a σ -algebra

1. $\emptyset \in F$
2. If $A \in F$, then $A^c \in F$
3. If $A_1, A_2, \dots \in F$ then $\bigcup_n A_n \in F$

Definition 0.1.2 (Probability Measure). A Probability measure on (Ω, F) is a function $\mathbb{P} : F \mapsto [0, 1]$ s.t.

- $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- If $A_1, A_2, \dots \in F$ is disjoint collection then $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(A_n)$

Exercise 0.1.3.

- $A, B \in F$

Proof. $A \cap B = (A^c \cup B^c)^c$

$A^c, B^c \in F \implies A^c \cup B^c \in F \implies (A^c \cup B^c)^c \in F$

□

- If $A, B \in \mathcal{F}, A \subset B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$
 $B = A \cup (B \setminus A) \implies \mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \implies \mathbb{P}(A) \leq \mathbb{P}(B)$

1 Events

Why Events

When three coins are tossed but not seen, we still take the sample space as:

$$\Omega = \{(b_1, b_2, b_3) \mid b_i \in \{H, T\}\}$$

After uncovering the first (or just after the first toss: is there a difference?)

$$F = \{\emptyset, \Omega, \{(HHH, (HTH), (HTT), (HHT))\}, \{(T, b_2, b_3) : b_i \in \{H, T\}\}\}$$

After uncovering the second,

$$\Omega_0 = \{H, T\}$$

$F = \{\emptyset, \Omega, \{H\} \times \{H\} \times \Omega_0\}$ plus other stuff and unions of stuff After uncovering the third, $F = P(\Omega)$

A glimpse of continuum

Let us model the height of a 'random' person

Alice uses a ruler in millimeters.

$$\Omega = \{\dots, \{1.7745, 1.775\}, \dots\}$$
 Event is what my rulers says

Bob uses a ruler in micrometers

$$\Omega = \{\dots, \{1.7777745\}, \dots\}$$

God uses a ruler

$$\Omega = \mathbb{R}[0, 3]$$

Don't do this set $\Omega = \mathbb{R}$ and let $F = thatstuff$

Terminology

Ω : Sample Space = all possible outcomes

(Ω, F) : Measurable space what you can know about what happened

$(\Omega, F, \mathbb{P})$: Probability Space how likely it is that an event happens

Note: If Ω is finite, one can put the uniform measure, i.e.

$$\forall \omega \in \Omega, \mathbb{P}(\{\omega\}) = \frac{1}{|\Omega|}$$

Properties

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- $\mathbb{P}(A) + \mathbb{P}(B) = \mathbb{P}(A \cup B) + \mathbb{P}(A \cap B)$

Example 1.0.1. Union bound: $\mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n)$

$$\bigcup_n A_n = A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus (A_2 \cup A_1)) \cup \dots$$

$$\mathbb{P}(\bigcup_n A_n) = \mathbb{P}(A_1) + \mathbb{P}(A_2 \setminus A_1) + \mathbb{P}(A_3 \setminus (A_2 \cup A_1)) + \dots$$

Since all the set minus is $\leq \mathbb{P}(A_n)$ thus $\mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n)$

Conditional Probability and Independence

Definition 1.0.2. $A, B \in F, \mathbb{P}(B) \neq 0$

The probability of A (conditionon/given) B is $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$

Definition 1.0.3. $A_1, A_2, \dots \in F$ and finite subcollection $i_1 < i_2 < \dots < i_n$

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_n}) = \mathbb{P}(A_{i_1}) \mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_n})$$

(A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$)

Example 1.0.4.

Independent weather/biased coin (70% H 30% T) toss. $\mathbb{P}(\text{rains}) = 70\%$

$$\Omega = \{(\text{rains}, H), (\text{rains}, T), (\text{clear}, H), (\text{clear}, T)\}$$

$$(\text{rains}) = \{(\text{rains}, H), (\text{rains}, T)\}$$

$$(\text{heads}) = \{(\text{rains}, H), (\text{clear}, H)\}$$

$$(\text{rains and heads}) = (\text{rains}) \cap (\text{heads})$$

Soothsayer's count toss: rain if heads, clear if tails

$$\Omega = \{(\text{rains}, H), (\text{clear}, T)\}$$

$$\mathbb{P}(\text{rains}) = 70\%$$

Independent and identically distributed (i.i.d) tosses:

$$n \text{ tosses, } \mathbb{P}(n\text{heads}) = 0.7^n$$

$$\mathbb{P}(\text{atleast 1tail}) = 1 - 0.7^n$$

Set Theorem $\longleftrightarrow$ Probability Theory

$$A \cap B \longleftrightarrow A \text{ and } B$$

$$A \cup B \longleftrightarrow A \text{ or } B$$

$$A^c \longleftrightarrow A \text{ does not happen}$$

$$A \setminus B \longleftrightarrow A \text{ happens } B \text{ does not happen}$$

Theorem 1.0.5 (Total Probability Theorem). $B_1, B_2, \dots, B_n$ with $\forall \mathbb{P}(B_i) > 0$, partitions Ω

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}((A \cap B_1) \cup \dots \cup (A \cap B_n)) \\ &= \sum_{i=1}^n \mathbb{P}(A \cap B_i) \\ &= \sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i)\end{aligned}$$

Example 1.0.6. 3 colors of products: 100 blue, 70 red, 30 purple

defective rate: 3% 9% 7%

partition $\Omega = B \cup R \cup P$

$$\mathbb{P}(B) = \frac{100}{200}, \mathbb{P}(R) = \frac{70}{200}, \mathbb{P}(P) = \frac{30}{200}$$

$$\mathbb{P}(D) = \mathbb{P}(D | B) \mathbb{P}(B) + \mathbb{P}(D | R) \mathbb{P}(R) + \mathbb{P}(D | P) \mathbb{P}(P)$$

$$\mathbb{P}(B | D) = \frac{\mathbb{P}(B \cap D)}{\mathbb{P}(D)} = \frac{\mathbb{P}(D|B)\mathbb{P}(B)}{\mathbb{P}(D)}$$

Condition Probability: rejection sampling (naive ver.)

Choose at random

- If defective
 - Make a histogram
- If not defective
 - Start again.

Example 1.0.7. A man possesses five coins, two of which are double-headed, one is double tailed, and two are normal. He shuts his eyes, picks a coin at random, and tosses it. What is the probability that the lower face of the coin is a head?

$$\mathbb{P}(DH) = 2/5, \mathbb{P}(DT) = 1/5, \mathbb{P}(N) = 2/5$$

$$\mathbb{P}(LH) = \mathbb{P}(LH | DH) \mathbb{P}(DH) + \mathbb{P}(LH | N) \mathbb{P}(N)$$

$$\mathbb{P}(LH) = \mathbb{P}(DH) + 1/2 \mathbb{P}(N) = 2/5 + 1/5 = 3/5$$

He opens his eyes and see that the coin is showing heads; what is the probability that the lower face is a head?

$$\mathbb{P}(LH | UH) = \frac{\mathbb{P}(LH \cap UH)}{\mathbb{P}(UH)} = \frac{\mathbb{P}(DH)}{\mathbb{P}(DH) + 1/2 \mathbb{P}(N)}$$

He shuts his eyes again, and tosses the coin again. What is the probability that the lower face is a head?

$$\mathbb{P}(LH' | UH) = \frac{\mathbb{P}(LH' \cap UH)}{\mathbb{P}(UH)} = \frac{\mathbb{P}(LH' \cap UH)\mathbb{P}(DH) + \mathbb{P}(LH' \cap UH|N)\mathbb{P}(N)}{\mathbb{P}(UH)}$$

He opens his eyes and see that the coin is showing heads; what is the probability that the lower face is a head?

$$\mathbb{P}(LH' | UH \cap UH') = \frac{\mathbb{P}(LH' \cap (UH \cap UH'))}{\mathbb{P}(UH \cap UH')}$$

He discards this coin, picks another at random, and tosses it. What is the probability that it shows heads?

Example 1.0.8. A random N of dice is thrown. Let A_i be the even that $N = i$, and assume that $\mathbb{P}(A_i) = 2^{-i}, i \geq 1$. Thus sume of the scores is S . Find the probability that:

$$\text{a. } N = 2 \text{ given that } S = 4. \mathbb{P}(N = 2 | S = 4) = \frac{\mathbb{P}(N=2 \cap S=4)}{\mathbb{P}(S=4)} = \frac{\mathbb{P}(S=4|N=2)\mathbb{P}(N=2)}{\mathbb{P}(S=4|N=1)\mathbb{P}(N=1) + \dots + \mathbb{P}(S=4|N=4)\mathbb{P}(N=4)} =$$

b. The largest number shown by an dice is r , where S is unknown

$$\mathbb{P}(M_r) = \mathbb{P}(M_r | N = 1)\mathbb{P}(N = 1) + \mathbb{P}(M_r | N = 2)\mathbb{P}(N = 2) + \dots$$

$$\mathbb{P}(M_r | N = n) = \frac{r^n - (r-1)^n}{6^n}$$

4! choices for first row for the first column there are 3! ways to finish it for the second row there are 3 spots remaining. but two have been picked for each spot. so there is 2! choices for two spots put them one for the reaming then 1 way for the remaining

$$4! \times 3! \times 4$$

Inclusion-Exclusion Priciple

Probability Version: may replace cardinality with $\mathbb{P}$

Random Variables

Definition 1.0.9. Given $(\Omega, F, \mathbb{P})$, a random variable X is a function $X : \Omega \rightarrow \mathbb{R}$ such that $\{X \leq a\} = \{\omega \in \Omega | X(\omega) \leq a\} \in F$ for any $a \in \mathbb{R}$

What we talk about when we talk about a random variable

(Cumulative) distribution function

$$F_X(a) = \mathbb{P}(X \leq a)$$

(Discrete) probability mass function

$$f_X(a) = \mathbb{P}(X = a)$$

Properties of Expectation

$\forall a, b \in \mathbb{R}$ and random variables X, Y then

$$E(aX + bY) = aE(X) + bE(Y)$$

if $\mathbb{P}(X \geq 0) = 1$, $E(X) \geq 0$

In general, $|E(x)| \leq E(|X|)$

If $A \in \mathcal{F}$ then $E(A) = \mathbb{P}(A)$

Example 1.0.10. Expected number of fixed points for a random permutation

$E()$